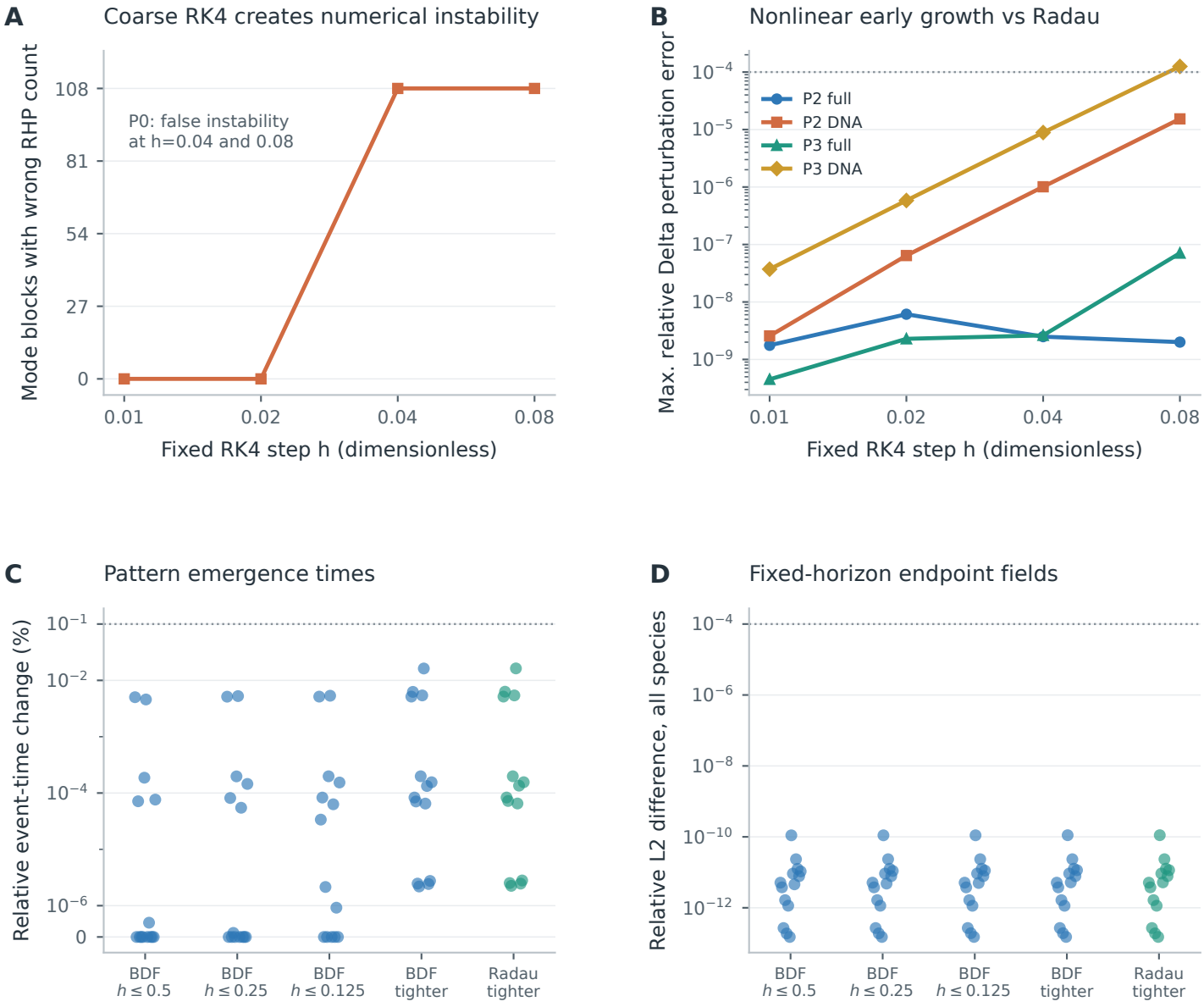
A
Coarse RK4 creates numerical instability
Mode blocks with wrong RHP count
108
81
54
27
0
P0: false instability
at h=0.04 and 0.08
0.01
0.02
0.04
0.08
Fixed RK4 step h (dimensionless)
B
Nonlinear early growth vs Radau
Max. relative Delta perturbation error
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
P2 full
P2 DNA
P3 full
P3 DNA
0.01
0.02
0.04
0.08
Fixed RK4 step h (dimensionless)
C
Pattern emergence times
Relative event-time change (%)
10^{-1}
10^{-2}
10^{-4}
10^{-6}
0
BDF
$h \le 0.5$
BDF
$h \le 0.25$
BDF
$h \le 0.125$
BDF
tighter
Radau
tighter
D
Fixed-horizon endpoint fields
Relative L2 difference, all species
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
BDF
$h \le 0.5$
BDF
$h \le 0.25$
BDF
$h \le 0.125$
BDF
tighter
Radau
tighter